

PARAPARAUMU, New Zealand

WMO: 934200

Lat: 40.9040S

Lon: 174.9840E

Elev: 26

StdP: 14.68

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.6	35.0	28.9	22.9	36.9	30.9	25.1	38.2	26.9	53.8	24.4	52.9	3.3	100	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.7	73.7	64.8	71.8	63.9	70.1	63.2	67.0	71.0	65.7	69.7	64.7	68.7	10.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	94.4	68.8	64.2	89.9	68.1	62.9	85.8	67.4	31.5	71.4	30.6	69.8	29.8	68.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.2	22.0	19.7		DB	28.6	78.5	1.3	3.1	27.7	80.8	26.9	82.6	26.2	84.3	25.3	86.6
				WB	27.7	69.1	1.3	1.7	26.8	70.3	26.0	71.3	25.3	72.3	24.3	73.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	62.8	63.1	60.6	56.6	53.6	49.4	47.9	49.0	51.7	54.0	56.5	60.6	(6)
	DBStd	6.73	3.90	4.06	4.33	4.68	4.91	5.02	4.65	4.18	4.22	3.88	3.79	3.89	(7)
	HDD50	317	0	0	0	4	22	74	97	72	33	13	2	0	(8)
	HDD65	3551	88	76	145	254	354	467	529	497	399	342	256	144	(9)
	CDD50	2301	397	366	328	201	133	57	34	40	84	136	197	328	(10)
	CDD65	59	20	23	8	1	0	0	0	0	0	0	0	7	(11)
	CDH74	51	19	16	7	0	0	0	0	0	0	0	0	9	(12)
	CDH80	2	0	1	0	0	0	0	0	0	0	0	0	1	(13)
(14) Wind	WSAvg	9.7	10.1	9.4	9.1	8.4	9.1	8.8	8.8	9.4	10.3	11.0	11.3	10.2	(14)
(15) Precipitation	PrecAvg	43.9	2.8	2.9	2.9	3.1	3.8	4.5	4.2	4.0	3.6	4.8	3.5	3.8	(15)
	PrecMax	60.4	5.2	15.3	8.3	7.1	7.7	6.8	9.7	6.7	9.1	14.0	7.2	6.7	(16)
	PrecMin	31.4	0.7	0.7	0.9	1.2	0.8	1.5	1.7	0.9	0.9	1.9	0.4	1.5	(17)
	PrecStd	6.4	1.2	2.9	1.5	1.6	1.8	1.5	1.9	1.6	1.7	2.2	2.0	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	76.1	74.6	69.7	65.8	61.6	59.5	59.7	62.2	65.5	70.3	74.4	(19)
		MCWB	66.7	65.2	64.6	62.5	59.4	56.9	54.3	53.8	55.2	58.0	61.7	64.2	(20)
	2%	DB	73.5	73.4	71.5	67.1	63.7	59.9	58.1	58.0	60.3	62.5	66.5	70.7	(21)
		MCWB	65.1	65.1	63.5	61.3	59.3	55.9	53.4	53.2	54.9	57.0	59.6	62.8	(22)
	5%	DB	71.3	71.7	69.4	65.4	62.2	58.6	56.8	56.8	59.1	61.1	64.3	68.7	(23)
		MCWB	63.9	64.3	62.5	60.1	58.1	54.8	52.5	52.5	54.6	56.0	58.3	62.0	(24)
	10%	DB	69.5	70.1	68.0	64.1	60.9	57.4	55.6	55.8	58.0	59.9	62.6	67.0	(25)
		MCWB	63.0	63.5	61.8	59.3	56.9	53.6	51.5	51.7	53.8	55.3	57.0	61.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	69.1	66.9	64.6	62.2	58.5	56.5	56.2	58.0	60.1	63.0	66.6	(27)
		MCDB	73.8	72.0	71.5	67.6	64.2	60.3	58.4	58.5	60.1	62.8	67.7	71.3	(28)
	2%	WB	66.8	67.3	65.3	62.8	60.3	56.8	54.7	54.6	56.6	58.3	61.2	64.7	(29)
		MCDB	71.0	70.8	69.4	65.6	62.6	59.1	57.0	56.9	58.9	61.2	65.0	68.4	(30)
	5%	WB	65.5	66.1	64.0	61.5	59.0	55.5	53.2	53.3	55.6	57.2	59.7	63.5	(31)
		MCDB	69.5	69.9	68.1	64.5	61.8	58.1	56.1	56.1	58.2	60.2	63.0	67.1	(32)
	10%	WB	64.2	64.8	62.8	60.1	57.5	54.1	52.0	52.2	54.5	56.2	58.3	62.3	(33)
		MCDB	68.4	68.9	67.1	63.5	60.5	57.0	55.1	55.3	57.5	59.3	61.5	65.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.7	13.1	13.0	11.6	11.8	12.4	11.8	10.9	10.5	10.9	10.8	(35)
		MCDBR	14.1	14.3	14.9	12.8	10.7	10.0	11.3	11.9	10.9	11.0	12.6	12.7	(36)
		MCWBR	8.3	8.6	9.3	9.0	8.2	8.2	8.8	9.0	8.1	7.9	8.5	7.6	(37)
	5% WB	MCDBR	11.2	10.7	11.6	9.8	8.9	8.5	9.3	9.7	8.7	9.0	9.9	9.8	(38)
		MCWBR	7.5	7.3	8.2	7.9	7.5	7.5	8.0	8.1	7.3	7.1	7.5	6.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.350	0.337	0.323	0.312	0.300	0.297	0.295	0.305	0.328	0.342	0.334	0.345	(40)
	taud		2.533	2.590	2.627	2.644	2.645	2.624	2.629	2.613	2.527	2.502	2.546	2.550	(41)
	Ebn at Noon		311	306	298	280	263	252	262	279	291	301	313	314	(42)
	Edh at Noon		34	31	27	23	20	18	20	23	30	33	34	34	(43)
(44) All-Sky Solar Radiation	RadAvg		2131	1874	1468	980	658	492	573	830	1203	1604	1993	2077	(44)
	RadStd		136	163	86	71	34	30	41	48	77	110	155	143	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page